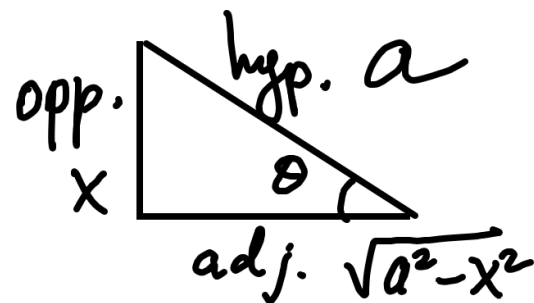


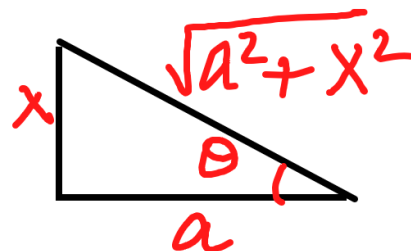
Trigonometric substitution

$$1) \sqrt{a^2 - x^2} \rightarrow x = a \sin(\theta) \\ dx = a \cos(\theta) d\theta$$

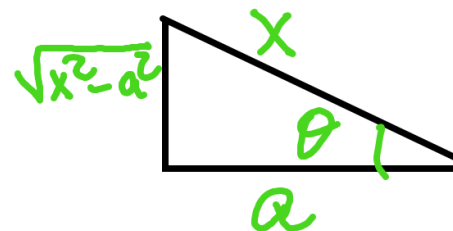


$$2) \sqrt{a^2 + x^2} \rightarrow x = a \cdot \tan \theta \\ dx = a \cdot \sec^2 \theta d\theta$$

$\tan \theta = \frac{x}{a} = \frac{\text{opp.}}{\text{adj.}}$



$$3) \sqrt{x^2 - a^2} \rightarrow x = a \sec \theta \\ dx = a \sec \theta \cdot \tan \theta d\theta$$



Examples:

$$1) \int \frac{\sqrt{9-x^2}}{x^2} dx$$

$$x = 3 \sin \theta$$
$$dx = 3 \cos \theta d\theta$$

$$> \int \frac{\sqrt{9 - (3 \sin \theta)^2}}{(3 \sin \theta)^2} \cdot 3 \cos \theta d\theta$$

$$= \int \frac{\sqrt{9 - 9 \cdot \sin^2 \theta}}{9 \cdot \sin^2 \theta} \cdot 3 \cos \theta d\theta = \int \frac{\sqrt{9(1 - \sin^2 \theta)}}{9 \cdot \sin^2 \theta} \cdot 3 \cos \theta d\theta$$

$$= \int \frac{\sqrt{9 \cdot \cos^2 \theta}}{9 \cdot \sin^2 \theta} \cdot 3 \cos \theta d\theta = \int \frac{\cancel{3} \cdot \cos \theta}{\cancel{9} \cdot \sin^2 \theta} \cdot \cancel{3} \cos \theta d\theta =$$

$$= \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \cot^2 \theta d\theta = \int \csc^2 \theta - 1 d\theta$$

$$= \int \csc^2 \theta d\theta - \int 1 d\theta = -\cot \theta - \theta + C$$

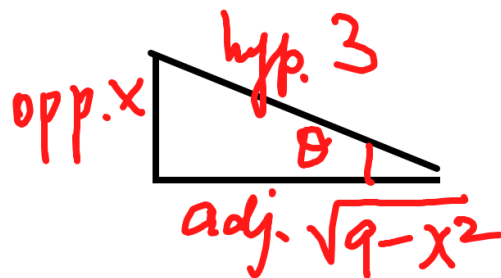
Thus, $-\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$

Since we let

$$x = 3 \sin \theta$$

$$\theta = \sin^{-1}\left(\frac{x}{3}\right)$$

$$\sin \theta = \frac{x}{3} \quad \begin{array}{l} \text{opp.} \\ \text{hyp.} \end{array}$$



$$\cot \theta = \frac{\text{adj.}}{\text{opp.}} = \frac{\sqrt{9-x^2}}{x}$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \rightarrow 1 + \tan^2 \theta &= \sec^2 \theta \\ \rightarrow \cot^2 \theta + 1 &= \csc^2 \theta \end{aligned}$$

$$2) \int \frac{1}{x^2 \sqrt{x^2+4}} dx$$

$$x = 2 \cdot \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$> \int \frac{1}{4 \cdot \tan^2 \theta \cdot \sqrt{4 \cdot \tan^2 \theta + 4}} \cdot 2 \cdot \sec^2 \theta d\theta$$

$$= \int \frac{2 \cdot \sec^2 \theta d\theta}{4 \tan^2 \theta \sqrt{4(\underbrace{\tan^2 \theta + 1}_{\sec^2 \theta})}} = \int \frac{2 \cdot \sec^2 \theta d\theta}{4 \cdot \tan^2 \theta \cdot \sqrt{4 \cdot \sec^2 \theta}}$$

$$= \int \frac{\cancel{2} \cdot \sec^2 \theta}{4 \cdot \tan^2 \theta \cdot \cancel{2} \cdot \sec \theta} d\theta = \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

Rewrite in terms of sine and cosine.

$$\frac{1}{\sin \theta} = \csc \theta$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1}{4} \int \frac{\frac{1}{\cos \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta}} d\theta = \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$\begin{matrix} u = \sin \theta \\ du = \cos \theta d\theta \end{matrix} > \frac{1}{4} \int \frac{1}{u^2} du = -\frac{1}{4u} + C$$



$$\frac{x}{2} = \tan \theta = \frac{\text{opp.}}{\text{adj.}}$$

$$\csc \theta = \frac{\text{hyp.}}{\text{opp.}} = \frac{\sqrt{x^2+4}}{x}$$

$$\text{Thus, } -\frac{1}{4} \frac{\sqrt{x^2+4}}{x} + C = -\frac{1}{4 \sin \theta} + C = -\frac{1}{4} \csc \theta + C$$

Memorize.

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

$$\int x^n dx = \frac{1}{-n+1} x^{-n+1} + C$$

$$\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$$

3) Evaluate

$$\int \frac{x}{\sqrt{3-2x-x^2}} dx = \int \frac{x}{\sqrt{4-(x+1)^2}} dx = \int \frac{u-1}{\sqrt{4-u^2}} du$$

This is an irreducible quadratic, so we bring it to the standard form by completing the square method.

Now u-sub.
 $x = u - 1 \leftarrow u = x + 1$
 $du = dx$

Now trig. sub. applies $\Rightarrow$

$$\begin{aligned} 3 - 2x - x^2 &= -x^2 - 2x + 3 \\ &= -\left[x^2 + 2x + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 - 3 \right] \\ &= -\left[(x+1)^2 - 4 \right] = -(x+1)^2 + 4 \\ &= 4 - (x+1)^2 \end{aligned}$$

$$= \int \frac{u-1}{\sqrt{4-u^2}} du = \int \frac{2 \cdot \sin \theta - 1}{\sqrt{4 - 4 \sin^2 \theta}} \cdot 2 \cos \theta d\theta$$

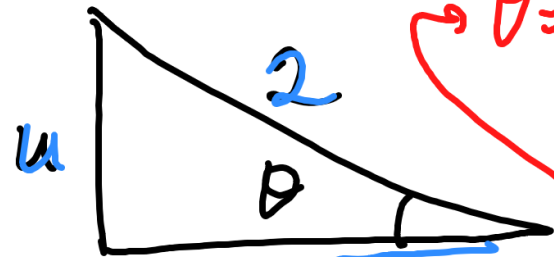
let $u = 2 \sin \theta$
 $du = 2 \cos \theta d\theta$

$4(1 - \sin^2 \theta) = 4 \cos^2 \theta$

$$= \int \frac{2 \sin \theta - 1}{\cancel{2 \cos \theta}} \cdot \cancel{2 \cos \theta} d\theta = \int 2 \sin \theta - 1 d\theta$$

$$= -2 \cos \theta - \theta + C$$

$$= -2 \cdot \frac{\sqrt{4 - (x+1)^2}}{2} - \sin^{-1}\left(\frac{x+1}{2}\right) + C$$



$\theta = \sin^{-1}\left(\frac{u}{2}\right)$

We let $\sin \theta = \frac{u}{2}$
 $\cos \theta = \frac{\sqrt{4-u^2}}{2}$

BUT $u = x+1$

Improper Integrals

→ **Type I** when the bound of integration is $\pm \infty$

OR both bounds are infinities, so we need to split the integral at $x = 0$

→ **Type II** when a function is undefined at one of the bounds of integration

In this case we need to rewrite a given integral with the one-sided limit

$$\int_2^{\boxed{5}} \frac{dx}{x-5} = \lim_{t \rightarrow 5^-} \int_2^t \frac{dx}{x-5}$$

$\nwarrow$
 $x \neq 5$

Make sure it is clearly state "-" or "+"!!!!

vs.

$$\int_{\boxed{2}}^5 \frac{dx}{x-2} = \lim_{t \rightarrow 2^+} \int_t^5 \frac{dx}{x-2}$$

→ **Type III** when a function is undefined at some value of x between the bounds of integration. At that value of x the integral must be split, so that we deal with two "sub-integrals" of type II.

$$\int_2^5 \frac{dx}{x-3} = \underbrace{\int_2^3 \frac{dx}{x-3}} + \underbrace{\int_3^5 \frac{dx}{x-3}}$$

Remark:

If either of these two "sub-integrals" diverges, then the original integral diverges.

The original integral converges ONLY if BOTH "sub-integral" are convergent.

Examples:

Evaluate integrals.

1) $\int_0^{\infty} 2x e^{-5x} dx$
type I

$$\rightarrow = \lim_{t \rightarrow \infty} \int_0^t 2x e^{-5x} dx = \lim_{t \rightarrow \infty} \left[-\frac{2x}{5} e^{-5x} - \frac{2}{25} e^{-5x} \right]_0^t \Rightarrow$$

Remark: Make sure you state LIMITS every time!!!

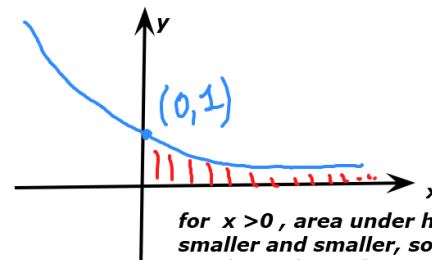
Integration by parts:

$$\begin{aligned} u &= 2x & dv &= e^{-5x} dx \\ du &= 2 dx & v &= -\frac{1}{5} e^{-5x} \end{aligned}$$

$$\int u dv = uv - \int v du = -\frac{2x}{5} e^{-5x} + \frac{2}{5} \int e^{-5x} dx = -\frac{2x}{5} e^{-5x} - \frac{2}{25} e^{-5x} + C$$

Hint: $y = e^{-x}$

this is an exponential decay function



for $x > 0$, area under the graph is getting smaller and smaller, so the integrals involving decay functions usually converge as $x \rightarrow \infty$

$$= \lim_{t \rightarrow \infty} \left(-\frac{2t}{5} e^{-5t} - \frac{2}{25} e^{-5t} \right) - \left(0 - \frac{2}{25} e^0 \right) = \boxed{\frac{2}{25}}$$

decays

1

This is a finite number, so this integral converges.

$$2) \int_2^5 \frac{dx}{x-5} = \lim_{t \rightarrow 5^-} \int_2^t \frac{1}{x-5} dx = \lim_{t \rightarrow 5^-} \ln|x-5| \Big|_2^t =$$

$$= \lim_{t \rightarrow 5^-} \ln|t-5| - \ln|2-5| = -\infty - \ln|-3| = \boxed{-\infty}$$

diverges

positive



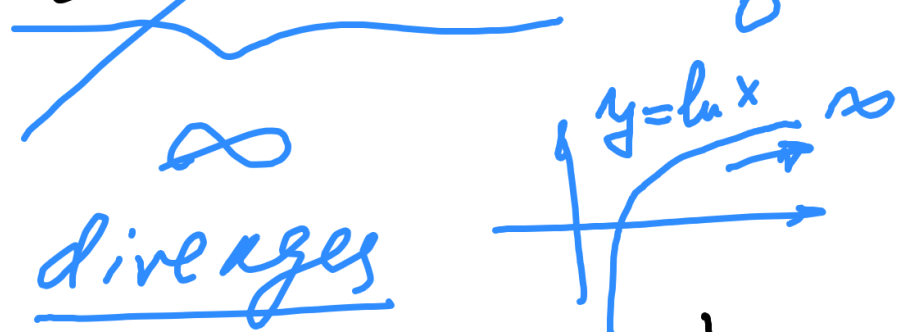
4.9, 4.99, 4.999, ..., -0.1, -0.01, -0.001, ...

$$3) \int_1^{\infty} \frac{\ln(x)}{x} dx = \lim_{t \rightarrow \infty} \frac{1}{2} (\ln x)^2 \Big|_1^t = \lim_{t \rightarrow \infty} \frac{1}{2} (\ln(t))^2 - \frac{1}{2} (\ln(1))^2$$

u-sub

$$u = \ln x$$

$$du = \frac{1}{x} dx > \int u du = \frac{1}{2} u^2 + C$$



$$4) \int_0^5 \frac{1}{\sqrt[3]{5-x}} dx = \lim_{t \rightarrow 5^-} \int_0^t \frac{1}{\sqrt[3]{5-x}} dx = \lim_{t \rightarrow 5^-} -\frac{3}{2} (5-x)^{2/3} \Big|_0^t =$$

u-sub

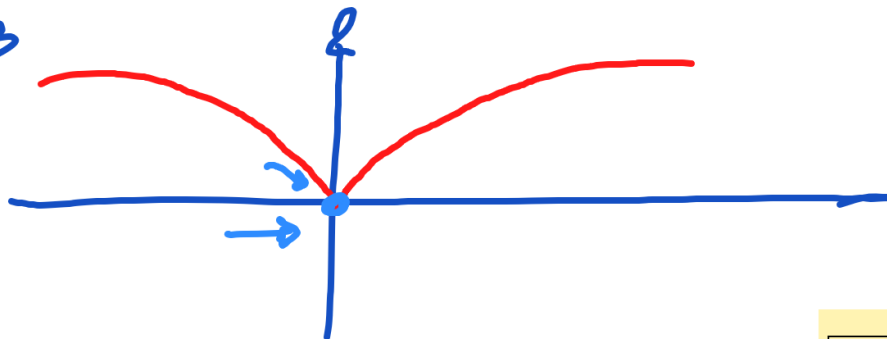
$$u = 5-x$$

$$-du = dx > -\int \frac{1}{u^{1/3}} du = -\int u^{-1/3} du = -\frac{3}{2} u^{2/3} + C$$

$$= \lim_{t \rightarrow 5^-} -\frac{3}{2} (5-t)^{2/3} + \frac{3}{2} (5)^{2/3} = \boxed{\frac{3}{2} \sqrt[3]{25}} \text{ finite converges}$$

Recall :

$$y = x^{2/3}$$



Comparison Test for Improper Integrals

→ If $f(x) \leq g(x)$ AND $\int_a^\infty g(x) dx$ CONVERGES, then $\int_a^\infty f(x) dx$ CONVERGES as well.

→ If $g(x) \leq f(x)$ AND $\int_a^\infty g(x) dx$ DIVERGES, then $\int_a^\infty f(x) dx$ DIVERGES as well.

Remark:

This test allows us to answer the question whether a given integral converges or diverges. Applying this test we can not determine the value to which the integral converges to, if it is convergent.

Recall :

$\int_a^\infty \frac{1}{x^p} dx$ $\left\{ \begin{array}{l} \rightarrow \text{if } p > 1, \text{ then integral converges} \\ \rightarrow \text{if } p \leq 1, \text{ then integral diverges} \end{array} \right.$

$$\int_0^a \frac{1}{x^p} dx = \begin{cases} \text{if } p < 1, \text{ then integral converges} \\ \text{if } p \geq 1, \text{ then integral diverges} \end{cases}$$

$$\int_a^\infty e^{-kx} dx \text{ converges for positive } k.$$

Examples:

Converges or diverges. Apply Comparison Test.

$$\int_1^{\infty} \frac{x+2}{\sqrt{x^3+1}} dx$$

To determine whether a given integral converges or diverges, we compare the highest exponents in the numerator and the denominator.

This is a divergent case.

We need to show that $g(x) \leq f(x)$

must have
smaller denominator

Solution:

for $x \geq 1$

$$x^3 + x^3 \geq x^3 + 1$$

$$(x+2) \cdot \frac{1}{\sqrt{x^3+x^3}} \leq \frac{1}{\sqrt{x^3+1}} \cdot (x+2)$$

$$g(x) = \frac{x}{\sqrt{x^3+x^3}} \leq \frac{x+2}{\sqrt{x^3+1}} = f(x)$$

AND

$$\int_1^{\infty} \frac{x}{\sqrt{x^3+x^3}} dx = \int_1^{\infty} \frac{1}{\sqrt{2}x^{1/2}} dx$$

Thus, the original integral diverges by comparison test

$p = \frac{1}{2} < 1$
diverges

$$\int_1^{\infty} \frac{1}{x^{1/2}} dx$$

$p = \frac{1}{2} < 1$
diverges

$$2) \int_1^{\infty} \frac{x^2 + 3x + 1}{x^6 + 2} dx$$

Thinking!

$$\boxed{k \int_1^{\infty} \frac{1}{x^p} dx}$$

as $x \rightarrow \infty$
Behaves as

$$\frac{x^2}{x^6} = \frac{1}{x^4} \quad \text{AND} \quad \int_1^{\infty} \frac{1}{x^4} dx$$

$p = 4 > 1$
converge

This is a convergent case

Need to show

$$f(x) \leq g(x)$$

for $x \geq 1$
denominator must be smaller

$$x^6 + 2 > x^6$$

$$(x^2 + 3x + 1) \cdot \frac{1}{x^6 + 2} < \frac{1}{x^6} \cdot (x^2 + 3x^2 + x^2)$$

$$\underbrace{\frac{x^2 + 3x + 1}{x^6 + 2}}_{f(x)} < \underbrace{\frac{x^2 + 3x^2 + x^2}{x^6}}_{g(x)} \quad \text{AND} \quad \Rightarrow$$

$$\int_1^{\infty} \frac{x^2 + 3x^2 + x^2}{x^6} dx = \int_1^{\infty} \frac{5x^2}{x^6} dx = 5 \int_1^{\infty} \frac{1}{x^4} dx$$

$p = 4 > 1$
converges.

Thus, $\int_1^{\infty} \frac{x^2 + 3x + 1}{x^6 + 2} dx$ converges as well by the Comparison test.

Recall: cosine is bounded between -1 and 1

$$3) \int_1^{\infty} \frac{3\cos(x) + 5}{\sqrt{x^3 + x}} dx$$

$p = 3/2 > 1$
converges

$$-1 \leq \cos(x) \leq 1$$

$$-3 \leq 3\cos(x) \leq 3$$

$$2 \leq 3\cos(x) + 5 \leq 8$$

for convergence
we use the green
box



for divergence we use the purple box

$$\frac{3 \cos(x) + 5}{\sqrt{x^3 + x}} \leq \frac{8}{\sqrt{x^3}} \quad \text{AND}$$

$$\int_1^{\infty} \frac{8}{\sqrt{x^3}} dx$$

$p = 3/2 > 1$
converges.

Thus, $\int_1^{\infty} \frac{3 \cos(x) + 5}{\sqrt{x^3 + x}} dx$ converges as well.

4) Which of the following integrals are convergent?

I $\int_2^{\infty} \frac{1}{(x + \sqrt{x})^{5/2}} dx$ $p = 5/2 > 1$ converges. IV

~~II~~ $\int_3^{\infty} \frac{1}{\sqrt{x-2}} dx$ $p = 1/2 < 1$ diverges. V

III $\int_2^5 \frac{1}{\sqrt{x-2}} dx$

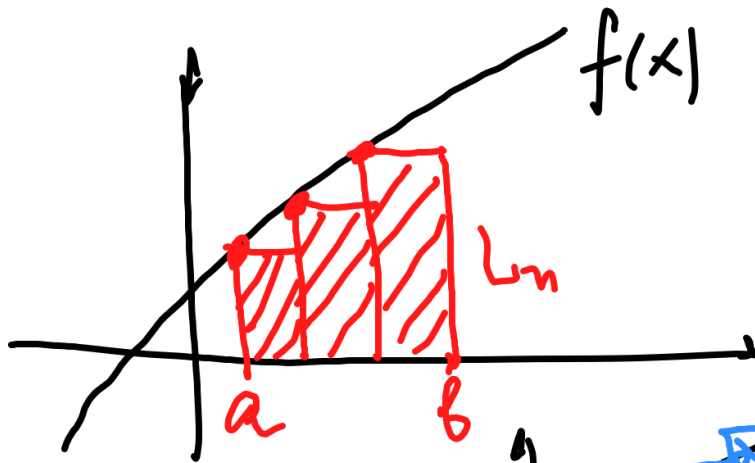
$\int_2^{\infty} \frac{1}{\ln(x)} dx$ $\frac{1}{x} < \frac{1}{\ln x}$
diverges

$\int_0^9 \frac{2}{\sqrt{x}} dx$ $p = 1/2 < 1$ converges.

Numerical Estimation of Definite Integrals

You need to know the graphs of functions, whether the graphs are increasing ($f' > 0$) or decreasing ($f' < 0$) or concave up ($f'' > 0$) or concave down ($f'' < 0$) on the given interval.

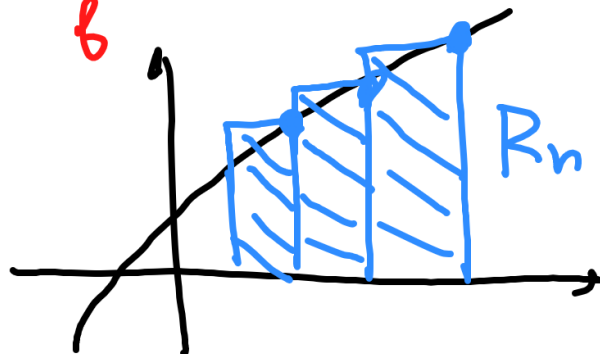
$$\text{LEFT}(n) = L_n \quad \text{vs.} \quad \text{RIGHT}(n) = R_n$$



Greater is the number of partitions on the interval $[a, b]$, more accurate is the estimated value of an integral.

if a function is increasing, the $L(n)$ underestimates an integral.

BUT the $\text{Right}(n)$ will overestimate.

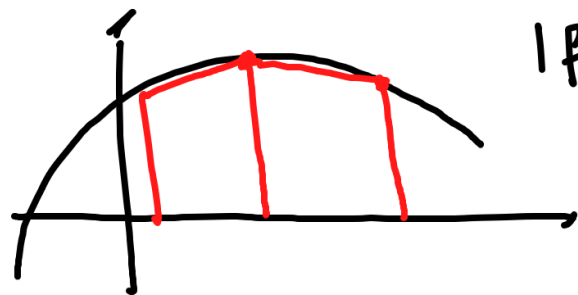


$$\text{LEFT}(n) < \int_2^{10} \ln(x) dx \leq \text{RIGHT}(n)$$

↑
Increasing

$$\text{LEFT}(n) \leq \text{LEFT}(20n) \leq \int_2^{10} \ln(x) dx \leq \text{RIGHT}(30n) \leq \text{RIGHT}(10n)$$

closer to the actual value if $n \rightarrow \infty$



If $f(x)$ is concave down on $[a, b]$ then

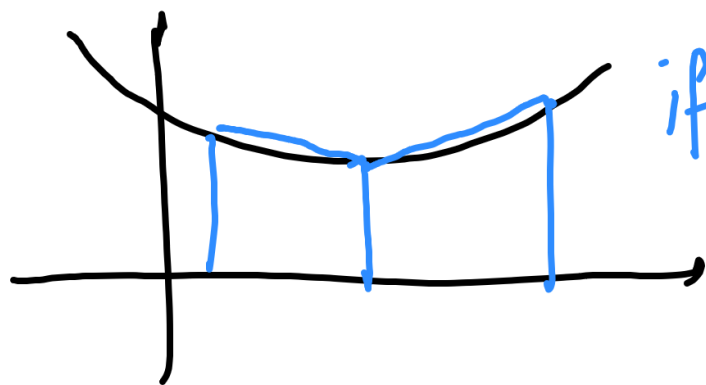
$$\text{TRAP}(n) \leq \int_a^b f(x) dx$$

underestimates

But

MID(n) will overestimate

$$\text{TRAP}(n) \leq \int_a^b f(x) dx \leq \text{MID}(n)$$

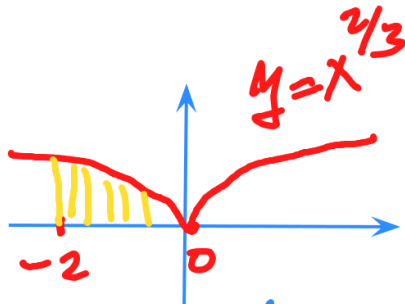


if $f(x)$ is concave up, then

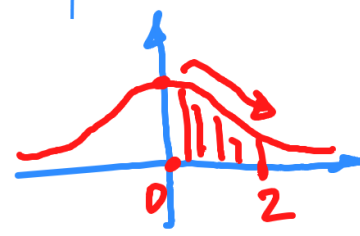
$$\text{MID}(n) \leq \int_a^b f(x) dx \leq \text{TRAP}(n)$$

True or False

(1) False $MID(n) \leq \int_{-2}^0 x^{2/3} dx \leq \underbrace{TRAP(n)}_{\text{underestimate}}$



(2) True $RIGHT(n) \leq \int_0^2 e^{-x^2} dx \leq LEFT(n)$



(3) False $LEFT(n) \leq \int_0^{10} \frac{1}{1+x^2} dx \leq RIGHT(n)$



(4) True
A midpoint rule will overestimate ✓

$\int_{\pi}^{2\pi} \tan^{-1}(x) dx$

$L(n)$ overestimates decreasing functions.

(5) True
A trapezoidal rule will underestimate

$\int_{\pi/2}^{\pi} \sin(x) dx$

